

Introduction

Basics of DC versus AC power and resistive versus reactive loads.

What is a Phasor?

Relationship between sinusoids and phasors. Deriving the concept of phasors. Formal phasor transformation and notation.

Life With and Without Phasors: Example Circuit Analysis

Seeing the benefits of phasors in practice over differential equations and the Laplace transform.

Complex Power and the Power Factor

Deriving complex power and the power triangle from time-domain instantaneous power. Power factor and power factor correction.

Three-Phase Power

Motivation for three-phase power. Wye versus delta configurations. Three-phase power in the grid.

Grid Modeling

Bus injection model. Power flow, optimal power flow, economic dispatch, and unit commitment problems.